# Hints: Functional sequences, series and spaces – 3. Power series

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** Cauchy–Hadamard is the root test: $\limsup \sqrt[n]{|a_n x^n|} = |x|/R$. For $|x| \leqslant r < R$ pick $q$ with $r/R < q < 1$; then $|a_n x^n| \leqslant q^n$ for large $n$, which also gives uniform convergence on $[-r, r]$ (M-test). For $|x| > R$ the terms do not even tend to 0.

**Hint 2.** The differentiated and integrated series have coefficients $na_n$ and $a_n/(n+1)$ (up to a shift of the index, which does not affect the radius); since $\sqrt[n]{n} \to 1$, the lim sup does not change.

**Hint 3.** Radii: for $\frac{n!}{n^n}$ use the ratio test, $\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e}$, so $R = e$. For $\sum x^{n^2}$ the nonzero coefficients are 1 at the positions $n^2$, so $R = 1$. For $\frac{x^n}{a^{\sqrt{n}}}$ $(a > 0)$, $a^{-\sqrt{n}/n} \to 1$, so $R = 1$. For $d(n)$, $1 \leqslant d(n) \leqslant n$ gives $R = 1$.

## Problem 2

**Hint 1.** Reduce to $\sum a_n = 0$ (subtract $a_0$). With $s_n = a_0 + \cdots + a_n$, Abel summation gives $f(x) = (1-x)\sum_n s_n x^n$ for $|x| < 1$. Since $(1-x)\sum x^n = 1$, split the sum at $N$ where $|s_n| < \varepsilon$ for $n > N$.

**Hint 2.** Apply it to $\ln(1+x) = \sum_{n \geqslant 1} \frac{(-1)^{n-1} x^n}{n}$ and $\arctan x = \sum_{n \geqslant 0} \frac{(-1)^n x^{2n+1}}{2n+1}$ on $(-1, 1)$: both series converge at $x = 1$ by Leibniz, and both functions are continuous at 1. For the converse: $\sum (-1)^n x^n = \frac{1}{1+x} \to \frac{1}{2}$, while $\sum (-1)^n$ diverges.

## Problem 3

**Hint 1.** Let $s_n = a_0 + \cdots + a_n$ and $\sigma_n = \frac{s_0 + \cdots + s_n}{n+1} \to \sigma$. Apply Abel summation twice: $\sum a_n x^n = (1-x)\sum s_n x^n = (1-x)^2 \sum (n+1)\sigma_n x^n$. (First check that the radius is at least 1, which follows from the second claim.)

**Hint 2.** Since $(1-x)^2 \sum (n+1) x^n = 1$, the right-hand side is a weighted average of the $\sigma_n$ with positive weights whose mass escapes to infinity as $x \to 1^-$; split at $N$ as in Abel's theorem to get $f(x) \to \sigma$.

**Hint 3.** For the second claim: $s_n = (n+1)\sigma_n - n\sigma_{n-1}$, so $\frac{s_n}{n} = \frac{n+1}{n}\sigma_n - \sigma_{n-1} \to \sigma - \sigma = 0$. In particular $a_n = s_n - s_{n-1} = o(n)$, so the radius is at least 1.

## Problem 4

**Hint 1.** Let $f(x) \to A$ and $na_n \to 0$. Estimate $s_N - f(x) = \sum_{n \leqslant N} a_n (1 - x^n) - \sum_{n > N} a_n x^n$ with the choice $x = 1 - \frac{1}{N}$, using $1 - x^n \leqslant n(1-x)$.

**Hint 2.** The first sum is at most $\frac{1}{N}\sum_{n \leqslant N} n|a_n|$, a Cesàro mean of a null sequence. The second is at most $\sup_{n > N} n|a_n| \cdot \frac{1}{N}\sum_n x^n = \sup_{n > N} n|a_n|$. Both tend to 0, and $f(1 - \frac{1}{N}) \to A$. Littlewood: $a_n = O(1/n)$ already suffices (much harder).

**Hint 3.** For any prescribed $c$: $\frac{1-x}{1+x} = 1 + 2\sum_{n \geqslant 1} (-1)^n x^n$ is a divergent series with Abel sum 0, and $2c\sum (-1)^n$ has Abel sum $c$ (if $c \neq 0$). Adding them gives a divergent series with Abel sum $c$ for every $c$.

## Problem 5

**Hint 1.** Fix $x_0$ in the interval and use Taylor's formula with integral remainder: $R_n(x) = \frac{(x-x_0)^{n+1}}{n!}\int_0^1 f^{(n+1)}(x_0 + s(x-x_0))(1-s)^n\,ds$. For $x > x_0$ all Taylor terms and the remainder are non-negative, so $0 \leqslant R_n(y) \leqslant f(y) - f(x_0)$ for every $y > x_0$ in the interval.

**Hint 2.** Since $f^{(n+1)} \geqslant 0$ has non-negative derivative, it is non-decreasing. For $x_0 < x < y$, comparing the integrands gives $R_n(x) \leqslant \left(\frac{x-x_0}{y-x_0}\right)^{n+1} R_n(y) \leqslant \left(\frac{x-x_0}{y-x_0}\right)^{n+1}\big(f(y) - f(x_0)\big) \to 0$.

**Hint 3.** So the Taylor series at $x_0$ converges to $f$ on $[x_0, y)$, and its radius is at least $y - x_0$. Given any point $c$ of the interval, choose $x_0 < c < y$: then $f$ coincides near $c$ with the sum of a power series centred at $x_0$, and such a sum is real-analytic at every interior point of its interval of convergence.